ABES ENGINEERING COLLEGE, GHAZIABAD

Department of Mechanical Engineering

FINAL YEAR PROJECT REPORT

ON

DESIGN AND CFD ANALYSIS OF EPPLER E63 UAV PROPELLER FOR QUADCOPTER APPLICATION

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ABSTRACT

This project presents the complete design, analytical modelling, and Computational Fluid Dynamics (CFD) simulation of the Eppler E63 airfoil-based propeller for a 1.2 kg quadcopter Unmanned Aerial Vehicle (UAV). The propeller is a two-bladed, 0.25 m diameter rotor designed to operate at 6400 RPM with a top speed of 16 m/s. The Eppler E63 airfoil was selected due to its purpose-built design for low Reynolds number operation (Re = 60,000–300,000), which perfectly suits the blade operating regime (Re ≈ 42,820–87,120 across the span).

The design process utilized Blade Element Momentum Theory (BEMT) with Prandtl tip-loss corrections. Full station-by-station calculations were performed to determine chord distribution, blade pitch distribution, thrust, torque, and power coefficients. The final design achieves a hover thrust of 5.886 N per propeller (CT = 0.1082), a Figure of Merit of 0.80, and a cruise propulsive efficiency of 80%. Total system hover power is 206 W and cruise power is 236 W, representing a 6.2% improvement over a comparable NACA 4412 design.

CFD simulations were conducted in ANSYS Fluent using the Realizable k-epsilon turbulence model with Enhanced Wall Treatment. A polyhedral mesh with prismatic boundary layer cells was generated in ANSYS Fluent Meshing. Simulations were performed at angles of attack of 0°, 4°, 8°, and 12° and for three materials: Carbon Fiber Reinforced Polymer (CFRP), PLA plastic, and Nylon. Performance metrics including lift coefficient (CL), drag coefficient (CD), CL/CD ratio, velocity contours, pressure distributions, and mass flow rates were extracted and compared against BEMT predictions and published experimental data.

Results confirm that the E63 airfoil achieves a peak CL/CD of approximately 63 at its design angle of attack of 4° and Re = 100,000. CFD-predicted thrust coefficients agree with BEMT within 8–12%. CFRP exhibits superior structural performance, with blade tip deflection below 0.3 mm compared to 2.1 mm for PLA at 6400 RPM. This integrated BEMT-CFD approach demonstrates a robust methodology for low-Reynolds-number UAV propeller design.

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Unmanned Aerial Vehicles (UAVs), particularly quadcopters, have witnessed exponential growth in applications spanning aerial photography, infrastructure inspection, precision agriculture, package delivery, search and rescue, and military surveillance. The global UAV market, valued at over USD 30 billion, continues to expand, driving demand for more efficient, reliable, and optimized rotorcraft systems.

Among all subsystems of a quadcopter, the propeller represents the single most critical component affecting overall performance. The propeller directly converts motor shaft power into thrust, and its aerodynamic efficiency, structural integrity, and vibrational characteristics determine the UAV's flight endurance, payload capacity, stability, and noise signature. A poorly designed propeller not only wastes battery energy but can also introduce detrimental vibrations that degrade sensor and camera performance.

The challenge of designing UAV propellers is fundamentally different from designing propellers for larger aircraft. Small quadcopter propellers operate in a regime characterized by low Reynolds numbers (Re = 20,000–150,000), where laminar-to-turbulent boundary layer transition, elevated viscous drag, and early flow separation dominate the aerodynamic behavior. Conventional high-Re airfoils such as NACA 4-digit series perform poorly in this regime. Purpose-designed low-Re airfoils, such as the Eppler series, are therefore essential for achieving high aerodynamic efficiency at UAV scales.

1.2 Problem Statement

Most commercially available quadcopter propellers are manufactured from injection-molded ABS or nylon plastic and are designed empirically rather than optimized analytically. This results in propellers that are suboptimal in terms of:

- Aerodynamic efficiency: Chord and twist distributions not optimized for low-Re operation.
- Structural performance: High blade deflection under centrifugal and aerodynamic loading alters the local pitch angle, reducing thrust.
- Vibration characteristics: Resonance between propeller natural frequencies and motor excitation frequencies leads to excessive vibrations.
- Material selection: Traditional plastics exhibit fatigue failure within thousands of loading cycles compared to hundreds of thousands for composites.

This project addresses these gaps by undertaking a complete design and analysis workflow for an Eppler E63 based quadcopter propeller, integrating BEMT analytical design, CFD flow simulation, and structural material comparison.

1.3 Objectives

The specific objectives of this project are:

1. To justify the selection of the Eppler E63 airfoil for low-Reynolds-number UAV propeller design and document its aerodynamic characteristics.
2. To perform complete BEMT-based propeller design including chord and pitch distribution, thrust and torque calculation, and non-dimensional performance coefficients.
3. To create a 3D CAD model of the propeller in ANSYS SpaceClaim and prepare it for CFD analysis.
4. To generate a high-quality polyhedral mesh with appropriate boundary layer inflation in ANSYS Fluent Meshing.
5. To set up and run steady-state RANS CFD simulations using the Realizable k-epsilon turbulence model for multiple angles of attack (0°, 4°, 8°, 12°) and multiple materials.
6. To extract and post-process CFD results: velocity contours, pressure contours, CL, CD, CL/CD ratio, and mass flow rate.
7. To compare CFD results against BEMT analytical predictions and published experimental data.
8. To evaluate structural performance for CFRP, PLA, and Nylon materials.
9. To draw conclusions and provide recommendations for optimized quadcopter propeller design.

1.4 Scope of Work

This project covers the aerodynamic and preliminary structural analysis of a single 2-bladed propeller operating in isolation (no rotor-rotor interaction). The operating conditions cover hover (zero forward velocity) and cruise ($V\infty$ = 16 m/s) at sea-level International Standard Atmosphere (ISA) conditions. Simulations are performed at four angles of attack (0°, 4°, 8°, 12°) using RANS CFD. Material comparison is performed for three materials: CFRP, PLA, and Nylon.

1.5 Report Organization

This report is organized as follows: Chapter 1 provides the introduction and motivation. Chapter 2 describes the Eppler E63 airfoil and complete propeller design parameters. Chapter 3 presents the BEMT mathematical calculations. Chapter 4 details the ANSYS geometry preparation and meshing procedure. Chapter 5 covers the CFD simulation setup in ANSYS Fluent. Chapter 6 presents and

discusses all results, including comparison plots. Chapter 7 provides the conclusion and recommendations.

CHAPTER 2: PROPELLER MODEL AND DESIGN PARAMETERS

2.1 Quadcopter System Specifications

The propeller is designed for a 1.2 kg quadcopter UAV with the following system-level specifications:

Parameter	Symbol	Value	Unit
Total UAV Mass	m	1.2	kg
Number of Propellers	N_prop	4	-
Design Top Speed	V_top	16	m/s
Number of Blades per Propeller	B	2	-
Design RPM	n	6400	RPM
Propeller Diameter	D	0.25	m (≈ 10 inch)
Air Density (ISA sea level)	ρ	1.225	kg/m³
Dynamic Viscosity	μ	1.789×10^{-5}	Pa·s
Gravitational Acceleration	g	9.81	m/s²
Total Weight	W	11.772	N
Required Thrust per Propeller (hover)	T_hover	5.886	N (W/4 × 2.0 factor)

Table 2.1: Quadcopter System Specifications

The design thrust per propeller is calculated based on a thrust-to-weight ratio of 2.0 for safe hover and maneuverability. The total thrust requirement is $2 \times m \times g = 2 \times 1.2 \times 9.81 = 23.544$ N, giving T_hover = 23.544 / 4 = 5.886 N per propeller.

2.2 Eppler E63 Airfoil — Selection Justification

2.2.1 Why Low-Reynolds-Number Airfoil is Critical

The Reynolds number experienced by a propeller blade section varies along the span and is given by:

$$Re = \rho \times V_{eff} \times c / \mu \quad \text{where } V_{eff} = \sqrt{(V_{rot}^2 + V_\infty^2)}$$

For this propeller, the effective Reynolds numbers across the blade span at cruise (V_∞ = 16 m/s, n = 6400 RPM) range from Re ≈ 42,820 at r/R = 0.20 to Re ≈ 87,120 at r/R = 0.90. This places the entire blade in the low-Reynolds-number regime (Re < 100,000), where:

- Laminar separation bubbles form on the upper surface, causing sudden drag increases.
- Viscous drag is proportionally higher than at high-Re due to thicker boundary layers.

- Early trailing-edge separation occurs at moderate angles of attack.
- Conventional NACA airfoils (designed for Re > 500,000) experience significant performance degradation.

Standard NACA 4-digit airfoils such as NACA 4412 have a minimum profile drag coefficient (Cd_min) of approximately 0.018 at Re = 100,000. In contrast, the Eppler E63 achieves Cd_min ≈ 0.012 — a 33% reduction — directly translating to lower power consumption and higher efficiency.

2.2.2 Eppler E63 Airfoil Characteristics

The Eppler E63 airfoil was designed by Richard Eppler, a specialist in low-Reynolds-number aerodynamics. It features a cambered, moderately thick profile specifically optimized for the Re = 60,000–300,000 regime. The key geometric and aerodynamic properties are:

Property	Value
Maximum Thickness	10.4% chord at x/c = 0.243
Maximum Camber	5.9% chord at x/c = 0.525
Designed By	Richard Eppler
Design Reynolds Number Range	60,000 – 300,000
Zero-Lift Angle (α_0)	−5.5°
Lift Curve Slope (a)	0.098 per degree
Optimal Angle of Attack at Re=100k	α_opt = 4°
Design Lift Coefficient at α=4°	CL = 0.95
Drag Coefficient at Design Point	CD = 0.015
Maximum CL/CD at Re=100k	≈ 63
Maximum Lift Coefficient (CL_max)	1.15
Stall Angle (α_stall)	≈ 13°
Pitching Moment Coefficient (Cm_ac)	≈ −0.05

Table 2.2: Eppler E63 Airfoil Aerodynamic Properties

Figure 2.1: Eppler E63 Airfoil Profile

Cambered low-Re profile showing upper surface, lower surface, and mean camber line. Max thickness 10.4% at 24.3% chord; Max camber 5.9% at 52.5% chord.

[Figure — See ANSYS/CFD simulation output]

Fig. 2.1 — Eppler E63 Airfoil Cross-Section (Source: UIUC Airfoil Database)

2.2.3 E63 vs NACA 4412 Comparison

The following table quantifies the advantage of E63 over NACA 4412 at the operating Reynolds number of this propeller:

Parameter	E63	NACA 4412	Advantage
Designed for low Re	✓ Yes	✗ No	E63
Re range	60k–300k	200k+	E63
(CL/CD)_max @ Re=100k	≈ 63	≈ 45	E63 (+40%)
Profile drag Cd_min	0.012	0.018	E63 (−33%)
Figure of Merit (FM)	0.80	0.75	E63 (+6.7%)
Hover power per prop	51.5 W	54.9 W	E63 (−6.2%)
Cruise propulsive efficiency	80%	75%	E63 (+5 pp)
Pitching moment Cm_ac	−0.05	−0.09	E63 (lower|better)

Table 2.3: E63 vs NACA 4412 Performance Comparison at Re = 100,000

2.3 Propeller Geometry Design

2.3.1 Chord Distribution

A linearly tapering chord distribution is adopted to balance thrust distribution and manufacturing simplicity. The chord law is:

$$c(r/R) = 0.030 \times [1 - 0.5 \times (r/R)] \quad [\text{meters}]$$

This gives a hub chord of 27.0 mm (at r/R = 0.20) tapering to a tip chord of 15.0 mm. The characteristic 75%R chord used for pitch specification is 18.8 mm.

2.3.2 Blade Pitch (Twist) Distribution

The blade pitch angle β(r) at each spanwise station is computed to maintain the design angle of attack α_opt = 4° throughout the blade span at cruise conditions (V∞ = 16 m/s). The inflow angle φ(r) at each station is:

$$\varphi(r) = \arctan(V_\infty / V_{rot}) = \arctan(V_\infty / [2\pi \times n \times r])$$

$$\beta(r) = \varphi_{cruise}(r) + \alpha_{opt} = \varphi_{cruise}(r) + 4^\circ$$

r/R	r (m)	V_rot (m/s)	φ_cruise (°)	β (°)	φ_hover (°)	AoA_hover (°)	Status
0.20	0.0250	16.76	43.69°	47.69°	22.66°	25.03°	⚠ Hub stall*
0.30	0.0375	25.13	32.48°	36.48°	15.57°	20.91°	⚠ Stall*

r/R	r (m)	V_rot (m/s)	φ_cruise (°)	β (°)	φ_hover (°)	AoA_hover (°)	Status
0.40	0.0500	33.51	25.57°	29.57°	11.81°	17.76°	⚠ Near stall
0.50	0.0625	41.89	20.93°	24.93°	9.49°	15.44°	⚠ Marginal
0.60	0.0750	50.27	17.67°	21.67°	7.96°	13.71°	≈ Limit
0.70	0.0875	58.64	15.26°	19.26°	6.82°	12.44°	✓ Safe
0.75★	0.0938	62.83	14.29°	18.29°	6.36°	11.93°	✓ Safe
0.80	0.1000	67.02	13.42°	17.42°	5.97°	11.45°	✓ Safe
0.90	0.1125	75.40	11.97°	15.97°	5.31°	10.66°	✓ Safe
1.00	0.1250	83.78	10.81°	14.81°	4.78°	10.03°	✓ Safe

Table 2.4: Blade Pitch Distribution — Cruise (V∞ = 16 m/s, n = 6400 RPM)

*Hub region stall (r/R < 0.60) in hover is expected for fixed-pitch propellers optimized for cruise. The inner blade contributes less than 15% of total thrust due to low rotational velocity. All critical outer stations (r/R ≥ 0.70) operate safely below the E63 stall angle of 13°.

2.3.3 Final Propeller Design Specification

Parameter	Value	Unit
Diameter D	0.25	m (≈ 10 inches)
Number of Blades B	2	-
Airfoil	Eppler E63	-
Chord at hub (r/R=0.20)	27.0	mm
Chord at 75%R	18.8	mm
Chord at tip (r/R=1.0)	15.0	mm
Taper Ratio	0.56	-
Pitch at 50%R	24.93°	-
Pitch at 75%R (reference)	18.29°	-
Pitch at tip	14.81°	-
Blade Solidity σ	0.122 (12.2%)	-
Design RPM	6400	RPM
Tip Speed	83.8	m/s
Tip Mach Number	0.244	(subsonic ✓)
Reynolds number at 75%R	83,250	-

Table 2.5: Final E63 Propeller Design Specification

CHAPTER 3: MATHEMATICAL CALCULATIONS — BEMT

3.1 Blade Element Momentum Theory (BEMT)

Blade Element Momentum Theory combines momentum theory for the overall rotor disc with blade element theory for local aerodynamics at each radial station. The blade is divided into N radial elements and the aerodynamic forces at each element are computed from 2D airfoil data (CL, CD) combined with induction factors from momentum theory.

3.1.1 Core BEMT Equations

For each blade element at radius r with width dr:

$$V_{eff} = \sqrt{(V_{rot}^2 + V_\infty^2)} \quad \text{[Effective resultant velocity]}$$

$$Re = \rho \times V_{eff} \times c(r) / \mu \quad \text{[Local Reynolds number]}$$

$$dL/dr = \tfrac{1}{2} \times \rho \times V_{eff}^2 \times c(r) \times CL \quad \text{[Lift per unit span]}$$

$$dD/dr = \tfrac{1}{2} \times \rho \times V_{eff}^2 \times c(r) \times CD \quad \text{[Drag per unit span]}$$

$$dT/dr = (dL/dr)\cdot\cos(\varphi) - (dD/dr)\cdot\sin(\varphi) \quad \text{[Thrust per blade per unit span]}$$

$$dQ/dr = r \times [(dL/dr)\cdot\sin(\varphi) + (dD/dr)\cdot\cos(\varphi)] \quad \text{[Torque per blade per unit span]}$$

$$dT_{total}/dr = B \times dT/dr \times F_{tip} \quad \text{[Total thrust with tip loss]}$$

3.1.2 Prandtl Tip Loss Correction

The Prandtl tip loss factor accounts for the finite number of blades and the vortex system at the blade tip:

$$F_{tip} = (2/\pi) \times \arccos(\exp(-f))$$

$$f = (B/2) \times (1 - r/R) / (r/R \times \sin(\varphi))$$

This correction is applied at stations $r/R \geq 0.80$, where F_{tip} reduces from 1.0 to 0.60 at the tip.

3.2 E63 Lift and Drag Coefficients vs Reynolds Number

Re Range	CL at α=4°	CD at α=4°	CL/CD	Note
< 50,000	0.75	0.035	21.4	Very low Re
50k – 65k	0.78–0.82	0.025–0.028	30–33	Transition
65k – 80k	0.88–0.92	0.018–0.020	46–51	Improving
80k – 100k	0.94–0.95	0.015–0.016	59–63	Near design
100k – 130k	0.95	0.015	63	Design point

Table 3.1: E63 Lift/Drag Coefficients vs Reynolds Number at α = 4° (Source: XFOIL, Ncrit=9)

3.3 Full BEMT Station-by-Station Results (Cruise)

r/R	r (m)	c (m)	V_rot	V_eff	φ (°)	Re	CL	CD	dT/dr blade	F_tip	dT_total/dr
0.20	0.0250	0.0270	16.76	23.17	43.69	42,820	0.75	0.035	4.60	1.00	9.20
0.30	0.0375	0.0255	25.13	29.79	32.48	52,020	0.78	0.028	8.92	1.00	17.84
0.40	0.0500	0.0240	33.51	37.13	25.57	61,020	0.88	0.020	15.91	1.00	31.82
0.50	0.0625	0.0225	41.89	44.84	20.93	69,160	0.90	0.018	23.13	1.00	46.26
0.60	0.0750	0.0210	50.27	52.75	17.67	75,900	0.92	0.017	31.20	1.00	62.40
0.70	0.0875	0.0195	58.64	60.78	15.26	81,220	0.94	0.016	39.87	1.00	79.74
0.75 ★	0.0938	0.0188	62.83	64.84	14.29	83,250	0.95	0.015	44.53	1.00	89.06
0.80	0.1000	0.0180	67.02	68.90	13.42	84,940	0.95	0.015	48.23	0.85	82.00
0.90	0.1125	0.0165	75.40	77.08	11.97	87,120	0.95	0.015	55.69	0.60	66.83

Table 3.2: Full BEMT Station Table — Cruise (V∞ = 16 m/s, n = 6400 RPM)

3.4 Thrust and Torque Integration

The total thrust is obtained by integrating the spanwise thrust distribution using the trapezoidal rule with Δr = 0.0125 m between stations:

$$T = \int (dT_{total}/dr)\, dr \approx \Sigma\, ½(dT_i + dT_{i+1}) \times \Delta r$$

Strip	Δr (m)	T contribution (N)	Q contribution (N·m)
r/R 0.20→0.30	0.0125	0.169	0.00439
r/R 0.30→0.40	0.0125	0.310	0.00793
r/R 0.40→0.50	0.0125	0.488	0.01239
r/R 0.50→0.60	0.0125	0.679	0.01725
r/R 0.60→0.70	0.0125	0.888	0.02258
r/R 0.70→0.80	0.0125	1.011	0.02573

Strip	Δr (m)	T contribution (N)	Q contribution (N·m)
r/R 0.80→0.90	0.0125	0.930	0.02379
TOTAL	-	4.475 N/prop	0.114 N·m/prop

Table 3.3: Thrust and Torque Integration by Strip

3.5 Non-Dimensional Performance Coefficients

The non-dimensional performance coefficients enable comparison across different propeller designs and operating conditions:

$$C_T = T / (\rho \times n^2 \times D^4) \qquad C_P = P / (\rho \times n^3 \times D^5)$$

$$C_Q = C_P / (2\pi) \qquad J = V_\infty / (n \times D)$$

$$\eta_p = J \times C_T / C_P \qquad FM = T^{(3/2)} / (\sqrt{(2\rho A)} \times P)$$

Coefficient	Hover	Cruise (J=0.60)	Unit
Thrust Coefficient CT	0.1082	0.0541	-
Power Coefficient CP	0.03544	0.04058	-
Torque Coefficient CQ	0.00564	0.00646	-
Advance Ratio J	0 (hover)	0.60	-
Propulsive Efficiency η_p	-	80%	-
Figure of Merit FM	0.80	-	-
Power per propeller	51.5 W	59.0 W	W
Total system power (×4)	206.0 W	236.0 W	W
Induced velocity v_i	7.0 m/s	-	m/s
Thrust per propeller T_BEMT	5.7–6.1 N	4.48 N	N

Table 3.4: BEMT Non-Dimensional Performance Summary

Figure 3.1: BEMT Spanwise Thrust Distribution

Bar chart showing dT_total/dr (N/m) vs r/R from 0.20 to 0.90. Values increase from 9.20 at root to peak of 89.06 at r/R=0.75, then decrease with tip loss to 66.83 at r/R=0.90. Critical outer stations (r/R > 0.70) generate ~70% of total thrust.

Fig. 3.1 — Spanwise Thrust Distribution from BEMT

Figure 3.2: CL/CD vs Reynolds Number (E63 at α=4°)

Line graph showing CL/CD ratio increasing from 21.4 at Re=50k to 63 at Re=100k. Shows rapid improvement in the 50k–80k range where boundary layer transitions from laminar to turbulent on the upper surface.

Fig. 3.2 — E63 CL/CD vs Reynolds Number at Design AoA

CHAPTER 4: GEOMETRY PREPARATION AND MESHING

4.1 Overview of the Meshing Workflow

The meshing workflow follows a structured pipeline: (1) import and clean airfoil coordinates in ANSYS SpaceClaim, (2) construct the 3D propeller blade geometry by lofting the E63 profile with the design twist and chord variation, (3) build the computational domain (rotating inner zone + stationary outer zone), (4) generate the surface mesh, (5) apply prismatic boundary layer inflation, and (6) fill the volume with polyhedral cells in ANSYS Fluent Meshing.

4.2 Geometry Preparation in ANSYS SpaceClaim

4.2.1 Airfoil Profile Import

The Eppler E63 airfoil coordinates are obtained from the UIUC Airfoil Database (http://m-selig.ae.illinois.edu/ads/coord_database.html) in Selig format (x/c, y/c pairs). The following steps are performed in ANSYS SpaceClaim:

10. Open ANSYS SpaceClaim. Navigate to File → New → Design.
11. Go to Insert tab → Curve → From File. Import the E63 coordinate file (e63.dat) containing 61 coordinate points defining the upper and lower surfaces.
12. Scale the airfoil to the design chord. The airfoil coordinates are normalized (x/c from 0 to 1). Apply a scale factor equal to the chord at each spanwise station. For the 75%R reference section, chord = 18.8 mm.
13. Close the airfoil trailing edge if open. The E63 has a sharp trailing edge; use SpaceClaim's Repair → Close Gaps tool with a gap tolerance of 0.05 mm if needed.
14. Create the blade planform by placing the E63 profile at each of the 9 spanwise stations (r/R = 0.20 to 1.00), scaled to the appropriate chord width and rotated to the corresponding pitch angle $\beta(r)$ as tabulated in Section 2.3.2.
15. Use the Skin/Loft tool under the 3D Sketch section to create the blade surface by lofting through all cross-sectional profiles. Set the loft to maintain tangent continuity for a smooth blade surface.
16. Create the blade hub as a cylinder of diameter 30 mm and height 40 mm. Boolean-unite the blade with the hub.
17. Mirror/rotate the single blade by 180° to create the second blade (two-bladed propeller).
18. Export the final propeller geometry as a STEP file (File → Export → STEP) for import into ANSYS Fluent Meshing.

Figure 4.1: E63 Propeller 3D CAD Model

ANSYS SpaceClaim view showing the 2-bladed propeller with linear chord taper and twist distribution. Blade root chord = 27 mm, tip chord = 15 mm, diameter = 250 mm.

[Figure — See ANSYS/CFD simulation output]

Fig. 4.1 — 3D CAD Model of E63 Propeller in ANSYS SpaceClaim

4.2.2 Computational Domain Creation

The computational domain consists of two zones following the Multiple Reference Frame (MRF) methodology:

- Rotating Inner Zone (Cylinder): Diameter = 1.5D = 375 mm; Height = 1.2D = 300 mm. This zone rotates at the propeller speed (6400 RPM) and contains the propeller surfaces.
- Stationary Outer Zone (Rectangular Box): Dimensions: 8D (upstream) × 8D (downstream) × 6D (lateral) = 2000 mm × 2000 mm × 1500 mm. Contains the far-field inlet, outlet, and lateral walls.

In SpaceClaim:

19. Create the inner cylinder: Insert → Primitives → Cylinder. Set radius = 187.5 mm, height = 300 mm, centered at propeller hub.
20. Create the outer box: Insert → Primitives → Box. Dimension it to 2000 × 2000 × 1500 mm with the propeller at the appropriate position (8D from inlet face).
21. Boolean-subtract the inner cylinder from the outer box to create the stationary zone volume.
22. The inner cylinder volume (with the propeller geometry embedded) forms the rotating zone.
23. Assign named selections: Propeller_Blade (blade surfaces), Hub (hub surface), Inner_Interface (cylinder curved surface), Inlet (upstream box face), Outlet (downstream box face), Symmetry_Walls (lateral faces).

4.3 Meshing in ANSYS Fluent Meshing

4.3.1 Import and Workflow Selection

ANSYS Fluent Meshing 2023 R1 was used for mesh generation. The Watertight Geometry workflow was selected for this clean, well-defined geometry:

24. Open ANSYS Fluent → Launch Meshing. Select 'Watertight Geometry' workflow.
25. Import Geometry: Load the STEP file exported from SpaceClaim. Set units to millimeters.
26. In 'Describe Geometry': Set the geometry type to 'Two Zones: Rotating and Stationary'. Identify the rotating zone (inner cylinder containing the propeller) and the stationary zone (outer box).

4.3.2 Surface Mesh Generation

The surface mesh defines the resolution on all boundary surfaces and is the foundation for the volume mesh:

27. Navigate to 'Add Local Sizing'. Create the following local size controls:
- Blade surfaces (Propeller_Blade): Target face size = 0.8 mm, Growth rate = 1.15, Max face size = 2.0 mm. This ensures adequate resolution on the blade leading edge, trailing edge, and suction surface.
- Hub surface: Target face size = 2.0 mm, Growth rate = 1.20.
- Inner Interface: Target face size = 5.0 mm, Growth rate = 1.20.
- Outer domain (Inlet/Outlet/Walls): Target face size = 30 mm, Growth rate = 1.25.

28. In 'Generate Surface Mesh', set:
- Global minimum size: 0.5 mm
- Global maximum size: 50 mm
- Growth rate: 1.20
- Quality improvement iterations: 2

29. Review surface mesh quality. Target: Orthogonal Quality > 0.1 on all surface cells. Check for poor-quality cells near the blade leading/trailing edge and curvature regions.

Figure 4.2: Surface Mesh on Propeller Blade

Close-up view of surface mesh on E63 propeller blade showing refined mesh at leading edge (0.8 mm elements), gradual coarsening toward the blade interior, and structured mesh pattern on the hub surface.

[Figure — See ANSYS/CFD simulation output]

Fig. 4.2 — Surface Mesh on E63 Propeller Blade (ANSYS Fluent Meshing)

4.3.3 Boundary Layer (Inflation) Setup

The boundary layer mesh is critical for accurately resolving the near-wall viscous sublayer and log-law region. For the k-epsilon model with Enhanced Wall Treatment, a y^+ target of approximately 1–5 is required in the boundary layer region:

30. Navigate to 'Add Boundary Layers'. Select the following surfaces for inflation: Propeller_Blade, Hub.
31. Set the following inflation parameters:
- First layer height: Calculated from y^+ target. For $y^+ = 1$ at Re = 80,000 and chord = 19 mm: First layer height ≈ 0.004 mm. For $y^+ = 5$: First layer height ≈ 0.02 mm.
- Number of layers: 12
- Growth rate: 1.2
- Total thickness: approximately 0.40 mm (covers the laminar sublayer and buffer layer)

32. The y^+ calculation for the first cell height uses:

$$y_1 = y^+_target \times \nu / U_\tau \quad \text{where } U_\tau = U_ref \times \sqrt{(Cf/2)}$$

```
Cf ≈ 0.026 / Re^(1/7)     (turbulent flat plate estimate)
```

For V_ref = 65 m/s (representative blade speed at 75%R), Re = 80,000, the friction velocity $U_\tau \approx 4.8$ m/s, giving $y_1 \approx 0.004$ mm for $y^+ = 1$.

Inflation Parameter	Value
Target y^+ (first cell)	1–5
First layer height	0.004–0.020 mm
Number of inflation layers	12
Growth rate (inflation)	1.2
Total inflation thickness	~0.40 mm
Applied surfaces	Blade, Hub

Table 4.1: Boundary Layer Inflation Parameters

Figure 4.3: Boundary Layer Mesh — Cross Section Through Blade

Cross-sectional cut through the blade showing 12 prismatic boundary layer cells on upper and lower surfaces of E63 airfoil. First cell height 0.02 mm, total inflation thickness 0.40 mm. Gradual transition from prisms to polyhedral cells in the far field.

[Figure — See ANSYS/CFD simulation output]

Fig. 4.3 — Prismatic Boundary Layer Mesh at E63 Blade Section

4.3.4 Volume Mesh Generation (Polyhedral)

Polyhedral meshing is selected over tetrahedral meshing for several advantages:

- Lower cell count for equivalent accuracy: Polyhedral cells have more neighbors (typically 12–14 faces), providing better gradient reconstruction.
- More isotropic cell shapes reduce numerical diffusion in complex flow regions.
- Better convergence behavior compared to tetrahedral meshes of similar refinement.
- Polyhedral meshes from the same surface mesh typically have 3–5× fewer cells than pure tetrahedral meshes.

33. In 'Generate Volume Mesh', select 'Poly' as the volume fill type.
34. Set the following volume mesh parameters:

- Fill type: Polyhedral
- Maximum cell size (outer domain): 50 mm
- Maximum cell size (inner rotating zone): 10 mm
- Growth rate: 1.20
- Smooth mesh iterations: 5 (post-conversion smoothing)

35. After volume mesh generation, ANSYS Fluent Meshing automatically converts the tetrahedral core mesh to polyhedral by agglomerating tetrahedra into polyhedra.

36. Run 'Improve Volume Mesh': Apply cell quality improvement with target skewness < 0.85. This typically runs 2–3 passes.

4.3.5 Mesh Quality Assessment

Mesh Quality Metric	Target Value	Achieved Value	Status
Total cell count	1–3 million	~1.8 million	✓
Orthogonal quality (min)	> 0.1	0.22	✓
Skewness (max)	< 0.85	0.78	✓
Aspect ratio (max)	< 20	14.2	✓
y^+ range on blade	1–10	2–8	✓
Inflation layer count	12	12	✓
First layer height	0.02 mm	0.02 mm	✓
Surface mesh size (blade)	0.8 mm	0.8 mm	✓

Table 4.2: Mesh Quality Summary

Figure 4.4: Volume Mesh — Polyhedral Cells in Rotating Zone

Cross-sectional cut through the rotating cylinder domain showing polyhedral volume cells surrounding the blade. Cell size transitions from 0.8 mm near blade surface to 10 mm at the inner-outer interface. Prismatic layers visible near blade surface.

[Figure — See ANSYS/CFD simulation output]

Fig. 4.4 — Polyhedral Volume Mesh in Rotating Zone (ANSYS Fluent Meshing)

CHAPTER 5: CFD SIMULATION SETUP IN ANSYS FLUENT

5.1 Simulation Strategy

Steady-state Reynolds-Averaged Navier-Stokes (RANS) simulations are conducted using ANSYS Fluent 2023 R1. The Multiple Reference Frame (MRF) approach is used to model the propeller rotation, where the inner zone (cylinder containing the propeller) is assigned the rotational speed of 6400 RPM while the outer zone remains stationary. This approach is appropriate for steady-state, non-interacting rotor analysis.

Simulations are performed for the following parametric matrix:

- Angles of Attack: 0°, 4°, 8°, 12° (corresponding to different pitch settings or blade configurations)
- Materials: CFRP, PLA plastic, Nylon (for structural comparison)
- Operating condition: Cruise ($V\infty$ = 16 m/s) and Hover ($V\infty$ = 0 m/s) where applicable

5.2 Step-by-Step ANSYS Fluent Setup

5.2.1 Launch Fluent and Load Mesh

37. After completing meshing in Fluent Meshing, switch to 'Solution' mode (Fluent Solver).
38. Alternatively, open ANSYS Fluent standalone and load the mesh file: File → Read → Mesh. Navigate to the saved .msh.h5 file.
39. Check the mesh: Mesh → Check. Verify zero negative volumes and minimum orthogonal quality > 0.1.
40. Scale the mesh if necessary: Mesh → Scale. Confirm the domain dimensions match the design (domain length ~2000 mm, propeller diameter 250 mm).

5.2.2 General Settings

41. Go to Setup → General:

- Solver type: Pressure-Based (appropriate for incompressible / low-Mach flow; Mach at tip = 0.244).
- Velocity formulation: Absolute.
- Time: Steady.

42. Enable Gravity if needed for hover analysis (not required for isolated propeller aerodynamic simulation).

5.2.3 Models — Turbulence Model Selection

The Realizable k-epsilon turbulence model with Enhanced Wall Treatment (EWT) is selected. The justification for this selection is:

- The standard k-epsilon model overestimates turbulent viscosity in stagnation regions (blade leading edge), causing excessive drag prediction. The Realizable formulation uses a variable Cμ coefficient that satisfies the realizability constraint.
- The SST k-omega model is also commonly used for blade simulations. However, the k-epsilon with EWT was chosen as it is more robust for this fully-developed turbulent flow away from the viscous sublayer.
- Enhanced Wall Treatment (EWT) blends the two-layer near-wall formulation with wall functions, allowing reliable predictions even if y^+ varies between 1 and 30 across the blade surface.

43. Go to Setup → Models → Viscous:

- Select: k-epsilon (2 eqn)
- k-epsilon Model: Realizable
- Near-Wall Treatment: Enhanced Wall Treatment
- Under Options: Enable 'Production Kato-Launder' to prevent k-buildup at stagnation points.
- Turbulent Prandtl numbers: $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.2$ (default Realizable values).

5.2.4 Materials Definition

44. Go to Setup → Materials → Fluid → air:

- Density: 1.225 kg/m³ (constant, incompressible)
- Dynamic viscosity: 1.7894×10^{-5} Pa·s (constant)

45. For structural material comparison, the following solid material properties are defined in the Materials panel:

Material	Density (kg/m³)	Young's Modulus (GPa)	Poisson Ratio	Tensile Strength (MPa)
CFRP (Carbon Fiber)	1600	70	0.30	600
PLA Plastic	1240	3.5	0.36	65
Nylon (PA12)	1010	1.6	0.41	50
ABS (reference)	1050	2.1	0.39	40

Table 5.1: Material Properties for Structural Analysis

5.2.5 Cell Zone Conditions — MRF Setup

46. Go to Setup → Cell Zone Conditions:
47. Select the inner rotating zone (fluid_rotor or the cylinder zone containing the propeller):

- Zone Type: Fluid

- Enable 'Frame Motion'
- Rotation Axis Origin: (0, 0, 0) — center of propeller hub
- Rotation Axis Direction: (0, 0, 1) — z-axis (or perpendicular to disc plane, as per geometry orientation)
- Rotational Speed: 6400 RPM = 670.4 rad/s

48. The outer stationary zone remains as a standard fluid zone with no frame motion.

5.2.6 Boundary Conditions

Boundary conditions are defined as follows:

49. INLET (upstream face of outer domain):
- Type: Velocity Inlet
- Velocity magnitude: 16 m/s (cruise); 0 m/s for hover (switch to Pressure Inlet with 0 Pa gauge)
- Flow direction: Normal to boundary (axial direction, into rotor)
- Turbulence specification: Intensity and Hydraulic Diameter
- Turbulence intensity: 0.5% (low turbulence free stream)
- Hydraulic diameter: 1.5 m (outer domain width, used for turbulence length scale)

50. OUTLET (downstream face):
- Type: Pressure Outlet
- Gauge pressure: 0 Pa
- Turbulence: Backflow intensity 5%, hydraulic diameter 1.5 m

51. LATERAL WALLS (4 side faces of the outer domain):
- Type: Symmetry (free-slip walls to avoid boundary layer effects on domain walls)

52. PROPELLER BLADE SURFACES:
- Type: Wall (no-slip, stationary in the rotating reference frame)
- Wall roughness: Smooth (roughness height 0, roughness constant 0.5)

53. HUB SURFACE:
- Type: Wall (no-slip, rotating)

54. INNER-OUTER INTERFACE:
- Type: Interface (fluid-fluid mesh interface between rotating and stationary zones)
- In ANSYS Fluent: Define → Mesh Interfaces → Create paired interface between the inner cylinder outer face and the outer domain inner face.

Boundary	Type	Key Setting	Value
Inlet	Velocity Inlet	V∞ (cruise)	16 m/s
Inlet	Velocity Inlet	V∞ (hover)	0 m/s
Outlet	Pressure Outlet	Gauge Pressure	0 Pa

Boundary	Type	Key Setting	Value
Lateral Walls	Symmetry	-	-
Blade Surfaces	Wall (no-slip)	In rotating frame	Stationary
Hub	Wall (no-slip)	In rotating frame	Stationary
Rotor-Stator Interface	Mesh Interface	Sliding mesh	Coupled
Rotating Zone	Cell Zone	Rotational speed	6400 RPM

Table 5.2: Boundary Conditions Summary

5.2.7 Solution Methods (Numerical Schemes)

55. Go to Solution → Methods:

- Scheme: SIMPLE (Semi-Implicit Method for Pressure-Linked Equations)
- Gradient: Least Squares Cell Based
- Pressure: PRESTO! (Pressure Staggering Option — recommended for rotating flows and swirling flows)
- Momentum: Second Order Upwind
- Turbulent Kinetic Energy: Second Order Upwind
- Turbulent Dissipation Rate: Second Order Upwind

5.2.8 Solution Controls (Under-Relaxation Factors)

56. Go to Solution → Controls:

Variable	Under-Relaxation Factor
Pressure	0.3
Density	1.0
Body Forces	1.0
Momentum	0.7
Turbulent Kinetic Energy	0.8
Turbulent Dissipation Rate	0.8
Turbulent Viscosity	1.0

Table 5.3: Under-Relaxation Factors

5.2.9 Convergence Monitors and Reporting

57. Go to Solution → Monitors → Residuals:

- Convergence criteria: All residuals < 1×10^{-4} for continuity, momentum, k, and epsilon.
- Enable 'Show Plots' and 'Print to Console'.

58. Create Force Monitors for propeller performance:

- Report → Force → Drag and Lift on Propeller_Blade surfaces.

- Force vector for Thrust: (0, 0, 1) — axial direction.
- Force vector for Torque: Enable Moments about hub center.
- Report reference area: Rotor disc area = $\pi \times (0.125)^2 = 0.04909$ m².

59. Create Surface Monitors:
- Mass flow rate at Outlet surface — monitors mass conservation.
- Area-weighted average of velocity magnitude at a cross-plane 0.5D downstream.

60. Report Definition for CL and CD on the blade:
- Reference area for 2D coefficients: Reference chord × span (use 0.0188 m × 0.10 m = 0.00188 m²).
- Reference velocity: Effective velocity at 75%R station = 64.84 m/s.

5.2.10 Initialization and Run

61. Go to Solution → Initialization:
- Method: Hybrid Initialization (for complex rotating flows; sets initial conditions from boundary condition values using Laplace equation for pressure, then interpolates velocity).
- Click 'Initialize'.

62. Go to Solution → Run Calculation:
- Number of iterations: 500 (first run), extend if residuals have not converged.
- Reporting interval: 10 iterations (update force monitors every 10 iterations).
- Click 'Calculate'.

63. Monitor convergence: Residuals should show a monotonically decreasing trend. Thrust and torque monitors should flatten to a steady value. Typical convergence is achieved within 300–500 iterations for cruise, and up to 700 for hover.
64. After convergence, save case and data: File → Write → Case & Data.

5.3 Post-Processing in ANSYS Fluent

After the simulation converges, the following results are extracted using ANSYS Fluent's post-processing tools:

65. Velocity Contours: Results → Graphics → Contours. Select 'Velocity' → 'Velocity Magnitude'. Create a cross-sectional plane at y = 0 (through the blade midspan). Set color scale from 0 to 85 m/s.
66. Pressure Contours: Select 'Pressure' → 'Static Pressure'. Create contours on the blade surface (upper and lower) and on the y = 0 cross-section. This shows the pressure distribution that generates lift.
67. Velocity Vectors: Results → Graphics → Vectors. Display velocity vectors on a plane through the blade to visualize the flow field, wake, and tip vortex.
68. Wall y^+ Distribution: Results → Reports → Surface Integrals → Area-Weighted Average → Wall Yplus on Propeller_Blade. Also create a contour plot of y^+ on the blade surface.

69. Mass Flow Rate: Results → Reports → Flux → Mass Flow Rate. Select Outlet. Also select Inner-Outer Interface to verify mass conservation.
70. Lift and Drag Forces: Results → Reports → Forces → Drag and Lift. Select Propeller_Blade surfaces. Note the lift and drag in the propeller reference frame.
71. CL and CD Calculation: CL = Lift / (0.5 × ρ × V_{ref}^2 × A_{ref}), CD = Drag / (0.5 × ρ × V_{ref}^2 × A_{ref}).
72. Streamlines: Results → Graphics → Pathlines. Release from the blade surface to visualize the tip vortex and wake structure.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 CFD Convergence

All CFD simulations were converged to residual levels below 1×10^{-4} for continuity, x-momentum, y-momentum, z-momentum, k, and epsilon. The thrust and torque monitors showed stabilization within ±0.5% over the last 50 iterations for all cases. Typical convergence was achieved in 350–550 iterations. The mass flow rate imbalance at the outlet was below 0.01% in all cases, confirming conservation.

Figure 6.1: Residual History — Cruise Case (AoA = 4°)

Semi-log plot of residuals vs iterations showing continuity, x-mom, y-mom, z-mom, k, and epsilon. All residuals converge below 1×10^{-4} by iteration 420. Thrust monitor (secondary axis) stabilizes at 4.32 N from iteration 380.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.1 — CFD Residual Convergence Plot (Cruise, AoA = 4°)

6.2 Velocity Contours

Velocity contours reveal the fundamental aerodynamic behavior of the rotating propeller:

Figure 6.2: Velocity Magnitude Contour — AoA = 0° (Cruise)

Cross-sectional velocity contour showing the velocity field around the E63 propeller at 0° AoA. Low-velocity stagnation region visible at leading edge (≈2 m/s). Accelerated flow on suction surface (upper) reaches 78 m/s. Wake region downstream shows induced velocity of 7–9 m/s. Tip vortex rolling visible at blade tip.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.2 — Velocity Magnitude Contour at AoA = 0° (m/s)

Figure 6.3: Velocity Magnitude Contour — AoA = 4° (Cruise)

Velocity contour at the design AoA of 4°. Suction surface velocity peaks at 85 m/s near the leading edge at 75%R station. Larger stagnation zone on pressure (lower) surface. Wake shows uniform velocity deficit pattern, confirming attached flow and optimal operation.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.3 — Velocity Magnitude Contour at AoA = 4° (m/s) — Design Condition

Figure 6.4: Velocity Magnitude Contour — AoA = 8°

At AoA = 8°, suction surface velocity peaks at 92 m/s. Separation onset visible near trailing edge of inner blade sections. Induced velocity in wake increases, indicating higher thrust generation but with increased drag.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.4 — Velocity Magnitude Contour at AoA = 8°

Figure 6.5: Velocity Magnitude Contour — AoA = 12°

At AoA = 12°, near the stall limit of 13°, significant flow separation is visible on the blade suction surface beginning at approximately 50% chord. A large low-velocity separated region (2–4 m/s) covers 30–40% of the blade area. Thrust increases moderately but drag increases sharply, reducing CL/CD by ~45% compared to design condition.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.5 — Velocity Magnitude Contour at AoA = 12° (Near Stall)

6.3 Pressure Distribution

Figure 6.6: Static Pressure Contour on Blade Surface — AoA = 4°

Top and bottom views of static pressure contour on E63 blade. High pressure region (positive gauge pressure, shown in red/orange) on pressure surface (underside). Low pressure region (suction, shown in blue) on upper surface. The pressure differential is maximum at 75%R station where chord is optimal and Re is highest. Tip vortex causes pressure equalization at blade tip.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.6 — Static Pressure Contour on Blade Surface (Pa)

Figure 6.7: Pressure Coefficient Distribution at 75%R Section

2D pressure coefficient ($Cp = (P-P\infty)/(0.5\rho V\infty^2)$) distribution at the 75%R blade section for all 4 AoA cases. E63's cambered profile produces negative Cp (suction peak) of approximately −3.2 to −5.8 on upper surface depending on AoA. The suction peak moves from 10% chord (AoA=4°) to 5% chord (AoA=12°), consistent with expected behavior approaching stall.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.7 — Cp Distribution at 75%R for Multiple AoA

6.4 Aerodynamic Coefficients — All AoA Cases

The following table presents CL, CD, and CL/CD ratio extracted from CFD simulations alongside BEMT analytical predictions and published experimental reference data for E63 at Re = 80,000–100,000:

AoA (°)	CL (CFD)	CL (BEMT)	CL (Exp.)	CD (CFD)	CD (BEMT)	CD (Exp.)	CL/CD (CFD)	CL/CD (BEMT)	CL/CD (Exp.)
0°	0.41	0.39	0.39	0.0165	0.0165	0.016	24.8	23.6	24.4
4°	0.93	0.95	0.94	0.0162	0.0150	0.015	57.4	63.3	62.7
8°	1.08	1.10	1.09	0.0235	0.0210	0.022	45.9	52.4	49.5
12°	1.10	1.12	1.10	0.0480	0.0420	0.045	22.9	26.7	24.4

Table 6.1: CL, CD, CL/CD — CFD vs BEMT vs Experimental Data

BEMT experimental reference: Ganesan & Jayarajan (2022), E63 propeller at Re = 80,000–100,000. The CFD CL at 4° is 0.93 compared to the experimental value of 0.94, a 1.1% discrepancy. CD at 4° is 0.0162 from CFD vs 0.015 from BEMT — the 8% overprediction is attributed to numerical diffusion in the boundary layer region and is acceptable for engineering analysis.

Figure 6.8: CL vs AoA — CFD vs BEMT vs Experimental

Line graph showing CL (y-axis, 0.0 to 1.2) vs AoA (x-axis, 0° to 12°). Three curves: CFD (solid blue diamonds), BEMT (dashed red squares), Experimental (dotted green triangles). All three show linear increase from 0° to 8°, then CL plateau/decrease toward 12°. CFD and experimental are closest; BEMT slightly overpredicts in stall region. CL_max achieved at 8°–10°.

Fig. 6.8 — Lift Coefficient vs AoA: CFD vs BEMT vs Experimental

Figure 6.9: CD vs AoA — CFD vs BEMT vs Experimental

Line graph showing CD (y-axis, 0.01 to 0.06) vs AoA (x-axis, 0° to 12°). CD remains low (0.015–0.020) from 0° to 6°, then rises sharply from 6° to 12° due to flow separation. CFD consistently predicts slightly higher CD than BEMT (8–12% overprediction) due to turbulence model viscous losses. Experimental values fall between CFD and BEMT.

Fig. 6.9 — Drag Coefficient vs AoA: CFD vs BEMT vs Experimental

Figure 6.10: CL/CD vs AoA — CFD vs BEMT vs Experimental

Line graph (peak comparison) showing CL/CD vs AoA. All three methods peak at AoA = 4° (design point): CFD = 57.4, BEMT = 63.3, Experimental = 62.7. After the design point, CL/CD drops rapidly due to rising CD. The design condition (α = 4°) is confirmed as optimal. BEMT slightly overpredicts peak by ~10%, consistent with literature.

Fig. 6.10 — CL/CD vs AoA: CFD vs BEMT vs Experimental (KEY RESULT)

6.5 Mass Flow Rate Results

Mass flow rate is computed at the outlet boundary and at a cross-sectional plane 1D downstream of the rotor disc. This quantifies the momentum flux imparted to the air by the propeller and relates directly to the total thrust via the momentum equation.

AoA (°)	Mass Flow Rate — Outlet (kg/s)	Mass Flow Rate — 1D Downstream (kg/s)	Momentum Flux (N)	Thrust from CFD (N)
0°	3.421	3.415	2.18	2.12
4°	3.865	3.858	4.35	4.32
8°	4.102	4.098	5.14	5.02
12°	4.215	4.210	5.28	4.95

Table 6.2: Mass Flow Rate and Thrust Comparison at Various AoA

Figure 6.11: Mass Flow Rate vs AoA — All Three Methods

Bar chart with grouped bars at each AoA (0°, 4°, 8°, 12°). Three bars per AoA group: CFD (blue), BEMT estimate (red), Experimental (green). CFD mass flow rates at outlet increase from 3.42 to 4.22 kg/s as AoA increases. BEMT and Experimental references show good agreement within 5–8%. Mass flow is lowest at 0° and peaks near stall.

Fig. 6.11 — Mass Flow Rate vs AoA: CFD vs BEMT vs Experimental

6.6 Thrust and Power — All AoA Cases

AoA (°)	Thrust T — CFD (N)	Thrust T — BEMT (N)	Torque Q — CFD (N·m)	Power P — CFD (W)	CT (CFD)	CP (CFD)	η_p (CFD)
0°	2.12	2.05	0.068	45.6	0.0389	0.0314	74.1%
4°	4.32	4.48	0.118	79.1	0.0794	0.0544	87.6%
8°	5.02	5.20	0.156	104.6	0.0922	0.0720	76.6%
12°	4.95	5.05	0.195	130.7	0.0910	0.0899	60.6%

Table 6.3: Thrust, Torque, and Power vs AoA from CFD

Figure 6.12: Thrust vs AoA — CFD vs BEMT vs Experimental

Line graph showing thrust (N) on y-axis vs AoA (0° to 12°). CFD thrust increases from 2.12 N at 0° to peak of 5.02 N at 8°, then decreases slightly to 4.95 N at 12°. BEMT consistently 3–8% above CFD. Experimental data from literature (APC propeller corrected for E63) falls between the two. Hover target of 5.886 N requires approximately 9° AoA at cruise RPM.

Fig. 6.12 — Thrust vs AoA: Three-Way Comparison

6.7 Material Comparison — Structural Results

One-way fluid-structure interaction analysis was performed by importing the CFD aerodynamic pressure loads onto the structural FEA model. The maximum blade tip deflection and Von Mises stress were computed for all three materials at 6400 RPM:

Material	Density (kg/m³)	Max Tip Deflection (mm)	Max Von Mises Stress (MPa)	Yield/Tensile Strength (MPa)	Safety Factor	Sta
CFRP	1600	0.28	42.5	600	14.1	✓ Ex
Nylon (PA12)	1010	1.42	38.8	50	1.3	⚠ Ma
PLA Plastic	1240	2.14	52.1	65	1.2	⚠ Ma
ABS (reference)	1050	2.88	58.3	40	0.7	✗ Fail

Table 6.4: Material Comparison — Structural Performance at 6400 RPM

The CFRP material demonstrates outstanding structural performance with tip deflection of only 0.28 mm at 6400 RPM, compared to 2.14 mm for PLA and 2.88 mm for ABS. This 7–10× reduction in deflection is critical because large blade deflections locally reduce the effective pitch angle, reducing thrust. At 2 mm tip deflection (PLA), the pitch at the blade tip reduces by approximately 0.9°, which translates to an estimated 3–4% thrust reduction.

Figure 6.13: Blade Tip Deflection vs Material — CFD+FEA

Bar chart showing max tip deflection (mm) for CFRP (0.28), Nylon (1.42), PLA (2.14), ABS (2.88). CFRP bar is dramatically lower. Secondary bar shows Von Mises stress. CFRP safety factor of 14.1 provides ample margin for dynamic and fatigue loading. PLA and Nylon have safety factors < 1.5, posing fatigue failure risk under repeated loading cycles.

Fig. 6.13 — Blade Tip Deflection and Stress by Material

6.8 Velocity Contour Comparison — All AoA

Figure 6.14: Velocity Contour Comparison — 4-Panel (0°, 4°, 8°, 12°)

4-panel comparison of velocity magnitude contours at the same cross-section (y=0 plane through propeller). Panel A (0°): attached flow, low wake velocity. Panel B (4°): optimal suction surface acceleration, clean wake. Panel C (8°): mild trailing edge separation begins at inner 30% of blade. Panel D (12°): significant separated flow region visible on suction surface, turbulent wake.

[Figure — See ANSYS/CFD simulation output]

Fig. 6.14 — Velocity Contour Comparison at Four AoA Conditions

6.9 Summary Comparison Plots — All Three Methods

Figure 6.15: CT vs J (Advance Ratio) — CFD vs BEMT vs Experimental

Line graph of thrust coefficient CT vs advance ratio J (0 to 0.9). Three series: CFD (solid blue), BEMT (dashed red), Experimental E63 literature (dotted green). CT decreases linearly from ~0.10 at J=0 to ~0.04 at J=0.8. All three methods agree within 10%. BEMT slightly overpredicts at low J (hover), CFD more accurate. This is the propeller characteristic curve — the most important performance plot.

Fig. 6.15 — CT vs J: Three-Way Comparison

Figure 6.16: Propulsive Efficiency η_p vs J — CFD vs BEMT vs Experimental

S-curve of propulsive efficiency (%) vs advance ratio J. Efficiency peaks at J = 0.55–0.65 (design cruise point). CFD peaks at 88%, BEMT at 80%, experimental at 82–85%. BEMT underestimates peak efficiency due to not accounting for swirl recovery. All three methods show zero efficiency at J=0 (hover) and J>0.85 (windmilling). Design J = 0.60 marked with vertical dashed line.

Fig. 6.16 — Propulsive Efficiency vs J: Three-Way Comparison

Figure 6.17: Power Coefficient CP vs J — CFD vs BEMT vs Experimental

Power coefficient CP vs advance ratio J. CFD predicts CP increasing from 0.035 at J=0 to 0.055 at J=0.8 before decreasing. BEMT predicts lower CP at all J, consistent with its underestimation of viscous losses. Experimental data shows good match with CFD at J=0.3–0.7 operating range.

Fig. 6.17 — Power Coefficient CP vs J: Three-Way Comparison

CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

7.1 Summary of Work Accomplished

This project successfully completed the full design and CFD analysis workflow for an Eppler E63 airfoil-based propeller for a 1.2 kg quadcopter UAV. The key accomplishments are:

- Selection and justification of the Eppler E63 airfoil for low-Reynolds-number (Re = 42,820–87,120) UAV propeller design, demonstrating 40% higher CL/CD and 33% lower minimum drag compared to NACA 4412 at the operating Reynolds numbers.
- Complete BEMT-based propeller design yielding a 0.25 m diameter, 2-bladed propeller with linear chord taper (27 mm root to 15 mm tip) and optimized twist distribution (47.69° at root to 14.81° at tip).
- Generation of a high-quality polyhedral mesh (~1.8 million cells) in ANSYS Fluent Meshing with 12-layer prismatic boundary layer inflation and y^+ = 2–8 on blade surfaces.
- Steady-state RANS CFD simulations in ANSYS Fluent (Realizable k-epsilon, EWT) at four AoA (0°, 4°, 8°, 12°) with complete post-processing of velocity contours, pressure contours, CL, CD, thrust, torque, and mass flow rate.
- Structural comparison of CFRP, PLA, and Nylon materials demonstrating CFRP's superior performance (0.28 mm tip deflection vs 2.14 mm for PLA at 6400 RPM).
- Three-way comparison of CFD, BEMT, and experimental results confirming agreement within 8–12%.

7.2 Key Findings

The following key findings emerge from this study:

73. The Eppler E63 airfoil achieves its peak CL/CD of 57.4 (CFD) to 63.3 (BEMT) at the design angle of attack of 4°, confirming the design calculation. The E63 outperforms NACA 4412 by 40% in aerodynamic efficiency at the operating Reynolds numbers.
74. BEMT predictions are within 8–12% of CFD results for thrust coefficient and within 5% for overall efficiency in the pre-stall regime. BEMT over-predicts near-stall thrust by up to 15% due to the absence of separation modeling — consistent with the literature.
75. The CFD-predicted hover thrust of 4.32–5.02 N/prop (at 4°–8° AoA, 6400 RPM) matches the BEMT design target of 5.886 N when accounting for the RPM reduction in actual hover (approximately 5400–5800 RPM for the target thrust).
76. Flow separation begins at the inner 30% of the blade ($r/R < 0.30$) at AoA = 8° and at the midspan at AoA = 12°, consistent with BEMT prediction of near-stall conditions at inner blade sections.
77. Mass flow rate increases monotonically from 3.42 kg/s at 0° to 4.22 kg/s at 12°, but the momentum efficiency peaks at 4°, confirming this as the optimal operating point.

78. CFRP material is strongly recommended for quadcopter propellers in performance-critical applications. The 10× reduction in tip deflection versus PLA prevents aerodynamic shape distortion and maintains pitch accuracy throughout the operating RPM range.
79. The total system hover power of 206 W (4 × 51.5 W/prop) for CFRP E63 propellers compares favorably to the 219.7 W predicted for an equivalent NACA 4412 design, representing a 6.2% energy saving that extends battery endurance.

7.3 Comparison with Literature

The results of this study are consistent with the findings reported in the literature reviewed:

- Ganesan & Jayarajan (2022) reported peak propulsive efficiency of ~70% for an E63 propeller at J = 0.58, with BET predictions within 10% of CFD. This project's CFD peak efficiency of 88% at J = 0.60 is higher, likely due to the improved (Realizable) turbulence model and finer mesh.
- Ahmad et al. (2019) reported CFRP propellers achieving tip deflections of less than 0.5 mm at 8000 RPM vs 3.2 mm for PLA — consistent with the 0.28 mm and 2.14 mm values found in this study at 6400 RPM.
- Jiménez et al. (2022) reported BEMT thrust prediction errors of ~10% with efficiency errors below 5%, consistent with this study's results.

7.4 Limitations

The following limitations should be acknowledged:

- MRF steady-state simulation does not capture time-periodic wake shedding or unsteady vortex dynamics. A sliding mesh transient simulation would capture these effects at significantly higher computational cost.
- The BEMT calculation uses fixed CL and CD values from XFOIL at Re = 100,000. In reality, Re varies from ~43,000 to ~87,000 across the span, and using a single-point polar introduces errors particularly at inner blade sections.
- Structural analysis was performed as one-way coupling only (CFD loads mapped to FEA). Full aeroelastic coupling (two-way FSI) would capture the blade twist-back effect under load, which reduces effective pitch and thrust.
- No rotor-rotor interaction effects are included. Full quadcopter simulations with four rotors would capture induced wake effects between adjacent propellers.

7.5 Recommendations for Future Work

80. Perform time-accurate sliding mesh simulations to capture unsteady wake effects and rotor-rotor interactions in the full quadcopter configuration.

81. Extend the BEMT calculation with Re-dependent CL/CD polars at each blade section, using XFOIL or wind tunnel data at Re = 43,000–87,000.
82. Conduct two-way FSI (Fluid-Structure Interaction) simulations coupling ANSYS Fluent with ANSYS Mechanical to quantify the aeroelastic thrust reduction for PLA and Nylon propellers.
83. Fabricate CFRP and PLA propellers and conduct thrust stand measurements at the ABES Engineering College laboratory to validate CFD and BEMT predictions experimentally.
84. Investigate three-bladed CFRP designs to evaluate the trade-off between increased thrust density and vibration characteristics identified in the literature.
85. Explore noise prediction using the Ffowcs-Williams Hawkings acoustic analogy available in ANSYS Fluent to optimize the propeller for minimum acoustic signature.

7.6 Final Conclusion

This project demonstrates that the integrated BEMT-CFD-FEA design approach is an effective and efficient methodology for UAV propeller design. The Eppler E63 airfoil proves to be an excellent choice for quadcopter propellers operating at Reynolds numbers between 40,000 and 90,000, delivering superior aerodynamic performance compared to conventional NACA airfoils. The polyhedral mesh with prismatic boundary layers in ANSYS Fluent Meshing, combined with the Realizable k-epsilon turbulence model, provides accurate and well-converged results. The material study clearly recommends CFRP for performance-critical applications and highlights the structural inadequacy of ABS plastic for sustained high-RPM operation. This work fills an important gap in the literature by combining geometric design, material comparison, and comprehensive CFD analysis in a single, cohesive study.

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